Quantum Optics Lab

Week 7

In this week's lab, you will be introduced to our quantum optics setup, which is based on the "Entanglement Demonstrator" from the company qutools. This system utilizes components you are now very familiar with -- a free running diode laser, mirrors, lenses, waveplates, and fibers -- as well as one new component, a non-linear crystal. With these components it produces two photons at a time that are in the entangled state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|H\rangle_1|H\rangle_2 + e^{i\phi}|V\rangle_1|V\rangle_2\right),$$

where $|H\rangle$ and $|V\rangle$ are the horizontal and vertical polarization state of the photons and the subscript refers to the direction the photon is traveling.

The goals for this experiment will be to:

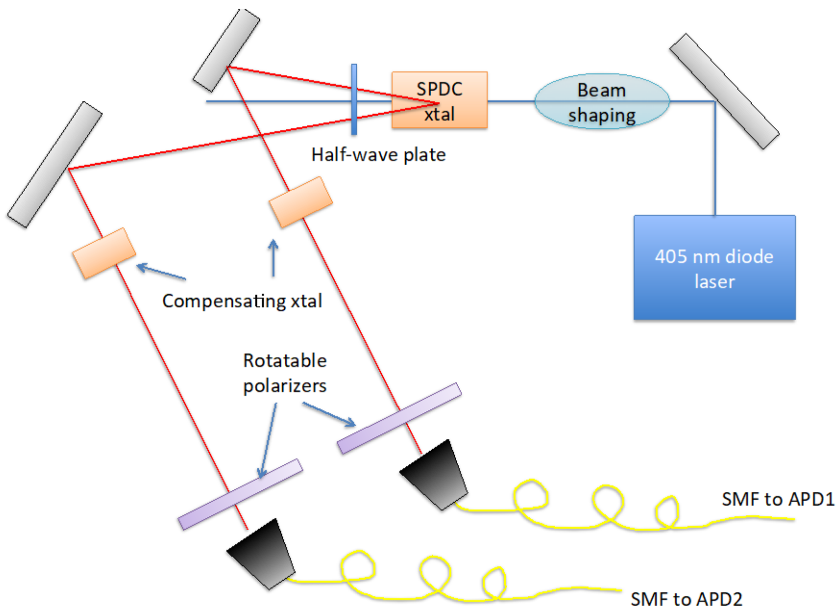
1. Learn about spontaneous parametric down conversion
2. Generate entangled photons
3. Verify that your photons are entangled.

Finally, we'll do a very famous experiment to measure something called "Bell's inequality." (really a closely related analog for polarizations called the "CHSH inequality"). This experiment not only confirms that the photons are entangled, but rules out a whole class of extensions to quantum mechanics. It proves that ψ really contains all the knowable information.

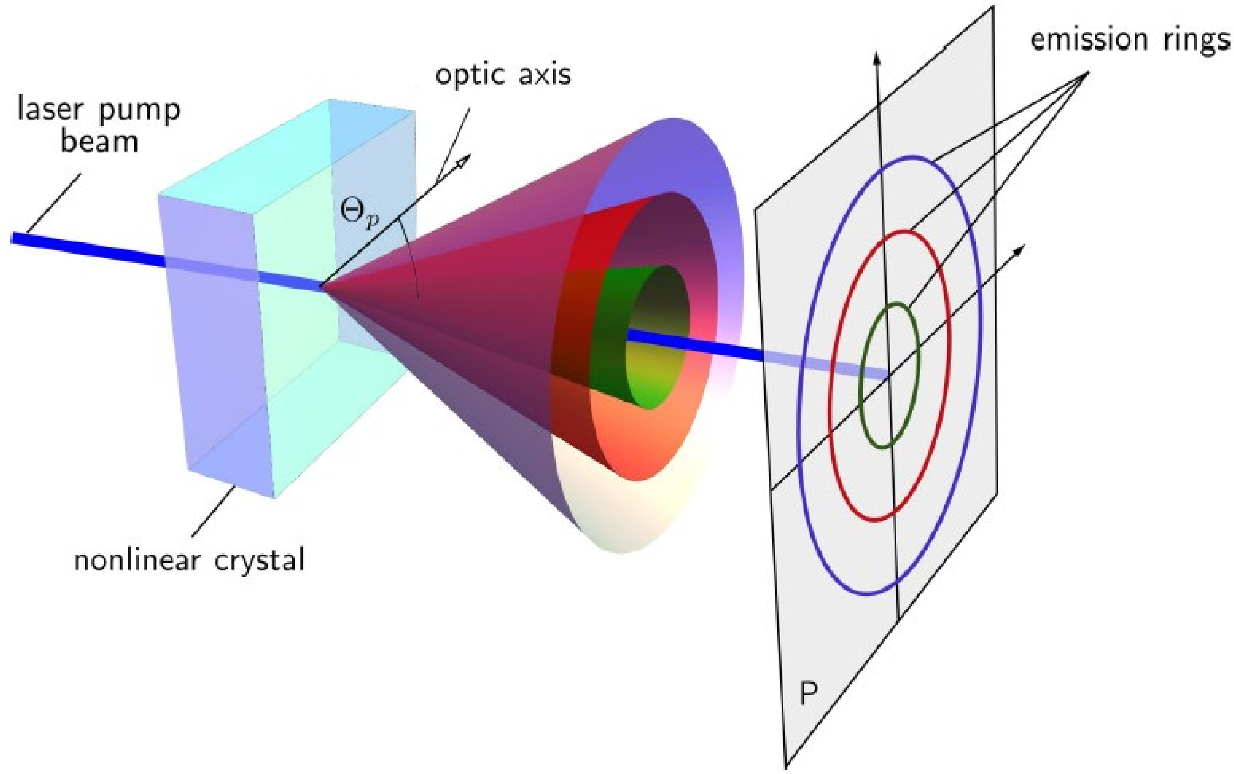
The Entanglement Demonstrator

The manual for both the "Entanglement Demonstrator" (aka the quED) and its control unit are both on the course website. Before proceeding with the rest of this document, stop what you're doing, download the manuals, and read them. This rest of this write-up will assume you have already read these manuals and are familiar with them.

The basic layout of the quED is as follows:

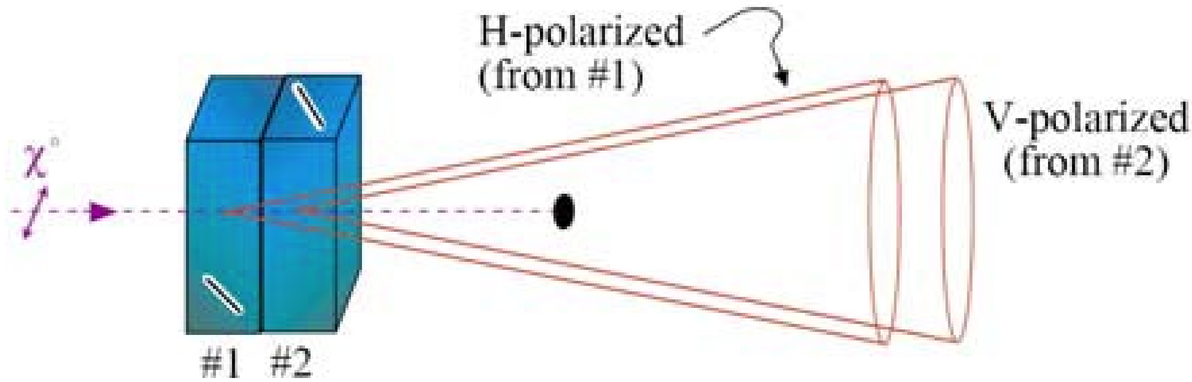


A 405 nm free-running laser diode (blue) is shaped and sent through two thin Beta-Barium-Borate (BBO) crystals that are essentially touching each other, where SPDC occurs. Because the SPDC process is inefficient, only about 1 blue photon in 10^{11} is down converted to two red photons. Thus, there are a very small number of down-converted, red photon pairs emerging from the crystals. The compensating crystals shown in the figure above may be absent from the apparatus you use.



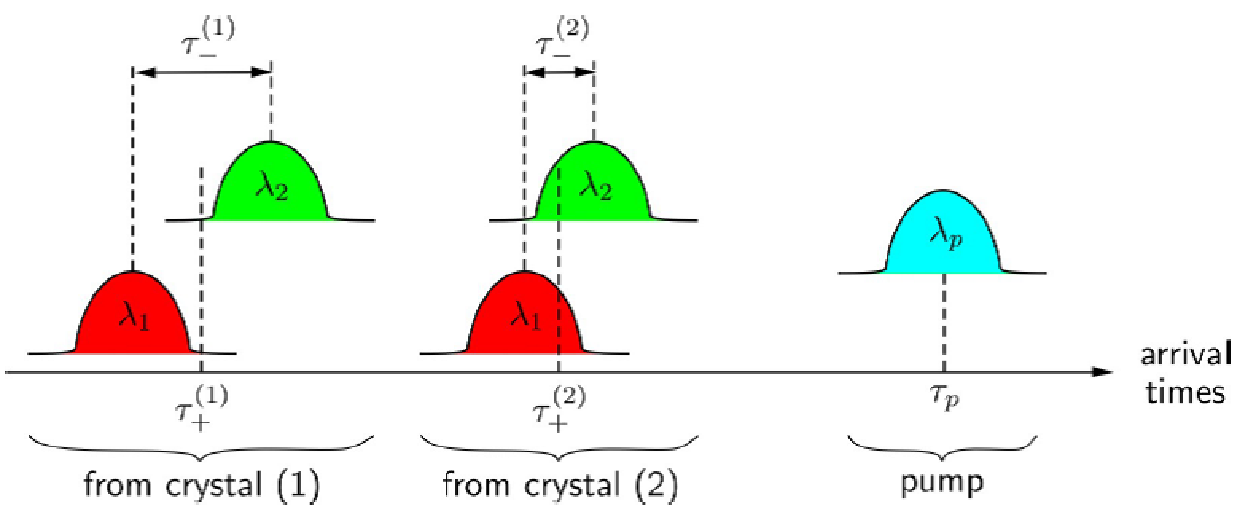
Because Type I phase matching is employed, the down converted photons come out of the first crystal or the second crystal, but these regions are so close together that they have about the same mode overlap with the fiber mode. Those created from the first crystal are H polarized, those created in the second are V polarized, but these locations in space are too close together to be resolved by the optical mode that couples into each fiber. So, when a blue photon is down-converted and leads to two red photons that emerge from the crystals and enter the fibers, it is impossible to know (even in principle) the polarization of the photon. However, we do know that they are perfectly correlated. Thus, the two-particle wavefunction (bipartite wavefunction) is entangled:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|H\rangle_1|H\rangle_2 + e^{i\phi}|V\rangle_1|V\rangle_2\right)$$



As you probably noticed in the manual, there is slightly more to the story than a couple of BBO crystals. The BBO crystals are birefringent – they actually have to be in order to get phase matching – so the horizontally and vertically polarized down converted photons and pump laser photons travel at different speeds and slightly different angles inside of the crystal. To compensate for this, a birefringent crystal is inserted into the pump beam to pre-compensate by adding the opposite delay between the H and V components of the pump beam.

Furthermore, since the two photons in Type I SPDC may have different wavelengths and two photons created in the first crystal have to travel through twice as much crystal, dispersion in the crystals leads to a separation of the photon wavepackets that is different for the H photons compared to the V photons. This would make it possible (in principle) to distinguish them and can destroy the entanglement.



To undo this effect, a dispersive, birefringent crystal is inserted after the BBO crystals to compensate for this effect.

Prelab

1. Using the quED II Inspection data sheet for either C3PO or QED-4 (our two quED devices in the lab) on the course website, calculate the number of photons/s in the pump laser beam when it is at its operating current. You may assume that the pump beam has a wavelength near the Blu-Ray scheme at 405 nm. If the total SPDC efficiency is 10^{-11}, what is the total rate of down converted photons do you expect? What is wavelength of the down converted photons? How much power is there in all of the down converted photons?
2. The control unit displays photon counts in each arm as well as the coincidence count rate. There are lots of noise processes that contribute to individual photons counts in each arm, however, the coincidence count rate (which requires both detectors to click) is fairly immune to these noise sources since noise in each detector is not correlated. Thus, whenever a photon is detected simultaneously in both arms, we can be quite sure it was from an SPDC event. The quED Inspection data sheet shows the coincidence count rate expected for our instrument. How does this compare to your calculation in problem 1? Explain the difference.
3. In lecture, we started with the "singlet" state $|\psi_s\rangle = \frac{1}{\sqrt{2}}(|H_1H_2\rangle - |V_1V_2\rangle)$. Calculate the probabilities for the four different measurement types for a given set of polarizer angles (*i.e.* $\langle\hat{\mathbf{A}}(\frac{\pi}{4})\hat{\mathbf{B}}(-\frac{\pi}{8})\rangle, \langle\hat{\mathbf{A}}(0)\hat{\mathbf{B}}(-\frac{\pi}{8})\rangle$, etc.), and use these to calculate the quantum mechanical expression for $\langle\hat{\mathbf{S}}\rangle$. Repeat this calculation using the "triplet" state $|\psi_t\rangle = \frac{1}{\sqrt{2}}(|H_1H_2\rangle + |V_1V_2\rangle)$. Does the expression for Bell's inequality need to be changed for the triplet state?
4. Rewrite $|\psi_{s,t}\rangle$ in the $|H(\alpha)\rangle, |V(\alpha)\rangle$ basis, where α is the angle between the polarizer's full transmission polarization orientation and the lab-frame horizontal direction.
5. Using the results of #4, write down the $|\psi_{s,t}\rangle$ states in the $\alpha = 45°$ basis. What does this mean and can you use it to develop an algorithm to ensure that you are making the desired Bell state?
6. OPTIONAL: An example hidden variable theory. Suppose that there was a hidden variable theory that purported to explain the results of our

polarization measurement. In this theory, whenever a pair of photons is created two $\lambda's$ (λ_1 and λ_2) are picked by some unknown underlying dynamics, which describes the outcome of the measurement of the photons' polarization. Let's suppose that these $\lambda's$ can be described as: $\lambda_1 = \theta$ and $\lambda_2 = \theta + \frac{\pi}{2}$, where θ is uniformly distributed between 0 and π. And the outcome of a measurement in the vertical basis is predicted from θ by:

$$P_V^{HVT}(\alpha,\lambda) = \begin{cases} 1, & |\alpha-\lambda| \leq \pi/4 \\ 1, & |\alpha-\lambda| > 3\pi/4 \\ 0, & \text{otherwise} \end{cases}$$

$$P_H^{HVT}(\alpha,\lambda) = \begin{cases} 1, & \pi/4 < |\alpha-\lambda| \leq 3\pi/4 \\ 0, & \text{otherwise} \end{cases}$$

That is, the photon polarization is measured to be V if λ is closer to the V axis than it is the H axis and vice versa. Using this information, calculate the expected correlation probabilities.(Hint: You should find that e.g. $P_{HV}^{HVT}(\alpha,\beta) = \frac{1}{2} - \frac{|\beta-\alpha|}{\pi}$)

7. OPTIONAL: Using the result of #6 calculate S. Does this theory violate the CHSH inequality?

Lab

1. **Before you begin the lab, find the TA or the professor. They will give you a tour of the apparatus. They will also ask you some basic questions about the apparatus that you must answer before you can begin. These questions will be based on information found in the manuals. Please come prepared by having already read the manuals thoroughly.**

2. **Spontaneous parametric down conversion.** Following the manual, turn on the pump laser diode and set its current at the operation value of 35 mA. Follow the manual procedure for optimizing the coincidence count rate.
    a. After the coincidence count rate has been optimized record the coincidence count rate as a function of the laser diode current between threshold (26 mA) and the operating current (35 mA). DO NOT EXCEED THE LASER DIODE OPERATING CURRENT OF 35 mA.
    b. Plot the coincidence rate vs. pump laser power. Do not measure the pump laser power, instead estimate it from the fact that threshold occurs at 27.4 mA and that at 35 mA it outputs 10 mW of power.
    c. Can you explain this graph? Recall the coupled equations we derived in class for the electric field amplitudes in SPDC. (Hint: just set all of the unknowns in those equations to 1 and see how the output power scales with input power.)
3. **Verifying entanglement Part I.** Go through the procedure in the "Verifying entanglement" part of the "Manual Bell Experiment" quED manual. Record all data and make the necessary plots. Report the visibility you calculated with this quick and approximate procedure. (We will do this right way in step #5).
4. **Improving entanglement.** Follow the "improving entanglement" section in the "Manual Bell Experiment" quED manual.
5. **Verifying entanglement Part II.** Do the next section in the "Manual Bell Experiment" quED manual. Record and plot:
    a. The coincidence count rate and the single photon count rate in each arm for each data point.
    b. Calculate the visibility. You should be able to achieve visibility of >97% in the horizontal/vertical basis and >90% in the diagonal basis.
    c. Does this prove the photons are entangled? If so, how? Does this rule out a local hidden variable explanation?
6. **Do the Bell experiment** (Section A.3 of the QuTools manual). Record the data. Be sure to include the accidental coincidence rates (the gray bars). Use your data to calculate $\langle \hat{S} \rangle$ and its uncertainty. Your value should be very close to that reported by the QuTools display.